No.	Parameter	Information
1	Full form	Simulation of Urban MObility
2	Type	Traffic simulation software
3	Developed by	German Aerospace Center (DLR)
4	Organization	DLR, Institute of Transportation Systems
5	Development started	2001
6	First open-source release	2002
7	Project ecosystem	Eclipse Foundation
8	License	Eclipse Public License 2.0
9	Cost	Free and open-source
10	Main purpose	Simulating traffic flow and analyzing congestion
11	Simulation type	Microscopic, with mesoscopic and macroscopic capabilities
12	Main applications	Traffic management, signal optimization, and route planning

Algorithm/model	Main purpose
Krauss model	Default vehicle-following model; maintains a safe distance between vehicles.
Intelligent Driver Model (IDM)	Models vehicle acceleration, braking, and following behaviour.
Wiedemann model	Simulates car-following behaviour.
Lane-changing models	Simulate vehicles changing lanes.
Dijkstra's algorithm	Calculates shortest paths in supported routing configurations.
A* algorithm	Calculates routes using heuristic-based path search.
Dynamic User Assignment (DUA)	Iteratively adjusts routes based on traffic conditions and travel costs.
Traffic signal control logic	Controls traffic-light phases using predefined or externally programmed rules.

Feature	Purpose
SUMO-GUI	Visualizes vehicles, roads, and traffic signals.
NetEdit	Creates and edits road networks.
TraCI	Connects Python to SUMO for retrieving data and controlling the simulation.
libsumo	Integrates SUMO functionality into an application without a separate TraCI connection.
Vehicle generation	Creates cars, bikes, trucks, and other vehicle types.
Traffic-light control	Supports fixed-time and custom signal control.
Vehicle detection	Provides vehicle information and supports simulated detectors.
Data export	Provides outputs for performance analysis.
Emergency vehicles	Supports emergency vehicle simulation.

Simulator	Developer/organization	License	Main use
SUMO	DLR / Eclipse SUMO project	Free and open-source	Traffic simulation and custom signal control
PTV Vissim	PTV Group	Commercial	Detailed traffic and intersection simulation
Aimsun Next	Aimsun	Commercial	Traffic planning and congestion analysis
CityFlow	CityFlow research project	Open-source	Efficient traffic simulation and AI-based signal control
MATSim	MATSim community	Open-source	Large-scale transportation and travel-demand simulation
TransModeler	Caliper Corporation	Commercial	Traffic engineering and transportation analysis
Paramics	Commercial transportation software ecosystem	Commercial; availability varies	Road traffic simulation

Type	Meaning	How it works	Example	Simulator
Microscopic	Individual vehicle simulation	Simulates each vehicle's movement, speed, acceleration, and interaction with other vehicles.	Tracking each car at a traffic signal.	SUMO, PTV Vissim
Macroscopic	Overall traffic flow simulation	Models traffic as a whole using traffic density, average speed, and flow.	Measuring the total traffic flow on a highway.	MATSim-related aggregate analyses; macroscopic models in Aimsun
Mesoscopic	Combination of microscopic and macroscopic approaches	Models traffic in groups or packets while representing some individual vehicle movements or interactions in a simplified way.	Analyzing traffic flow across a large city network.	Aimsun Next, SUMO's mesoscopic simulation mode

Parameter	Information
Full form	Traffic Control Interface
What is TraCI?	An interface that connects external programs with a running SUMO simulation.
Developed by	SUMO development team and contributors.
Organization	German Aerospace Center (DLR) / Eclipse SUMO project
Introduced	Documented in research publications in 2008.
Main language used	Python
Other supported languages	C++, Java, .NET, and MATLAB interfaces
Communication method	TCP-based client-server communication
Main purpose	Retrieves simulation data and controls vehicles and traffic signals.
Uses in our project	Counts vehicles, obtains traffic information, and changes traffic-light states through Python.
Main alternative	**libsumo**
Other alternative	**libtraci**, a C++ client library compatible with the libsumo API
Advantage	Enables dynamic traffic control during simulation.
Limitation	Communication overhead can slow simulations.

Parameter	Information	
Feature	**TraCI**	**libsumo**
Connection	Separate client-server connection	Runs SUMO functionality within the application
Communication	Uses socket communication	Avoids socket communication overhead
Performance	May have more overhead	Generally faster for suitable workloads
Python support	Yes	Yes
Best suited for	Flexible external control and GUI-based simulation	Performance-sensitive simulations

Library used in project :

No.	Library	Type	Use in Our Project
1	traci	SUMO library	Connects Python with SUMO, counts vehicles, retrieves vehicle information, and controls traffic signals.
2	sumolib	SUMO library	Not imported in the provided code.
3	math	Python built-in	Calculates distances, rounds timer values upward, and performs mathematical operations.
4	os	Python built-in	Handles file paths, checks environment variables, and locates SUMO files.
5	random	Python built-in	Generates random vehicle types, routes, arrival times, and vehicle colours.
6	time	Python built-in	Measures real-world elapsed time between vehicle horn sounds.
7	xml.etree.ElementTree	Python built-in	Creates XML route files, defines vehicle types, and reads SUMO network configuration.
8	pygame	External library	Plays traffic sounds, vehicle horns, ambulance sirens, and fire-truck sirens. It is imported inside the code when sound is enabled.